WATER : SCIENCE, SOCIETY & POLICY

ENVS/SDP 251 Spring Semester 2025

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Density

- **Definition**: Density (ρ) is the mass per unit volume of a substance.$\rho=mV$For water, the density is approximately **1 g/cm³ (1000 kg/m³)** at room temperature.

- **Importance**: Water's density plays a critical role in phenomena like buoyancy and stratification in oceans and lakes. The anomalous behavior of water (highest density at 4°C) is crucial for aquatic life.

Polarity and Hydrogen Bonding

- **Polarity**: Water is a **polar molecule** because of its bent shape and the difference in electronegativity between oxygen and hydrogen. The oxygen atom has a partial negative charge, and the hydrogen atoms have partial positive charges.
- **Hydrogen Bonding**: The polarity allows water molecules to form **hydrogen bonds** with each other, creating a highly cohesive structure.
- **Importance**: Hydrogen bonding is responsible for many of water's unique properties, such as its high boiling point, surface tension, and ability to dissolve ionic and polar substances.

Cohesion, Adhesion, and Surface Tension

- **Cohesion**: The attraction between water molecules due to hydrogen bonding.
    - Example: Water droplets forming a bead on a surface.
- **Adhesion**: The attraction between water molecules and other substances.
    - Example: Water climbing up a glass or a plant's xylem.
- **Surface Tension**: The cohesive forces at the surface of water create a “skin-like” layer, allowing small insects to walk on water.
    - **Explanation**: Surface tension is a result of the unbalanced hydrogen bonding at the surface.

Specific Heat of Water

- **Definition**: The specific heat of water is the amount of energy required to raise the temperature of 1 gram of water by 1°C.c=4.18 J/g°C
- **Explanation**: Water’s high specific heat enables it to absorb and store large amounts of heat, moderating Earth’s climate and maintaining stable aquatic environments.

Latent Heat of Vaporization

- **Definition**: The energy required to convert 1 gram of water from liquid to gas without changing its temperature.Latent Heat=2260 J/gLatent Heat=2260J/g**Explanation**: Water’s high latent heat makes it effective at moderating temperature through processes like sweating and evaporation, as these processes absorb significant heat from the environment.

Thank you for you time and attention